

Yinuo Du

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Summary

My research is to study human and AI decision-making in **cybersecurity**. Toward this goal, I create **computational models of humans**, design **AI Agents** for autonomous defense, and **complementary human-AI** teaming paradigms. My work involves **game theory**, **reinforcement learning**, **cognitive modeling**, and **behavioral experiments**.

Academic Employment

2025 – now  **Assistant Professor of Research**, Computer Science Department, University of Texas, El Paso.

Education

2021 – 2025  **Ph.D. Societal Computing**, School of Computer Science, Carnegie Mellon University.
Advisor: *Cleotilde Gonzalez, Fei Fang*
Committee: *Cleotilde Gonzalez, Fei Fang, Christian Lebiere, Prashanth Rajivan, Tiffany Bao*
Thesis topic: *Human and AI Decision-Making in Cybersecurity: A Multiagent Modeling Perspective*.

2019 – 2021  **M.Sc. Information Technology**, Information Networking Institute, Carnegie Mellon University.

2015 – 2019  **B.S. Software Engineering**, Xi'an Jiaotong University.

Honors and Awards

2025  **SCS Presidential Fellowship**, Carnegie Mellon University

2023-2024  **Women in Cybersecurity Student Scholarship**

2023  **Accelerating Foundation Models Research**, PI: Cleotilde Gonzalez,
To advance the development and application of foundation models in AI

2018  **Foho Technical Innovation Grants**,
For research excellence and academic performance, 1 of 77 awarded

2016-2018  **National Encouragement Scholarship**, Xi'an Jiaotong University.
For strong academic performance, 3 of 77 awarded

Research Publications

Journal Articles

- 1 **Yinuo Du**, Palvi Aggarwal, Kuldeep Singh, Fei Fang, and Cleotilde Gonzalez (2025a). "A Model of Human Behavior in Group Prisoner's Dilemma Games". In: *Psychological Review*. Under Review.
- 2 **Yinuo Du**, Palvi Aggarwal, Kuldeep Singh, Fei Fang, and Cleotilde Gonzalez (2025b). "Emergent Cooperative Behavior in Group Prisoner's Dilemma Games". In: *Acta Psychologica*.
- 3 **Yinuo Du**, Baptiste Prebot, Tyler Malloy, Fei Fang, and Cleotilde Gonzalez (2024). "Experimental Evaluation of Cognitive Agents for Collaboration in Human-Autonomy Cyber Defense Teams". In: *Computers in Human Behavior: Artificial Human*.
- 4 **Yinuo Du**, Baptiste Prebot, Tyler Malloy, and Cleotilde Gonzalez (2024). "A Cyber-War Between Bots: Cognitive Attackers are More Challenging for Defenders than Strategic Attackers". In: *ACM Transactions of Social Computing*.
- 5 Prebot, Baptiste, **Yinuo Du**, and Cleotilde Gonzalez (2023). "Learning about simulated adversaries from human defenders using interactive cyber-defense games". In: *Journal of Cybersecurity* 9.1.

Books and Chapters

- 1 Miah, Mohammad Sujan, Palvi Aggarwal, Marcus Gutierrez, Omkar Thakoor, **Yinuo Du**, Oscar Veliz, Kuldeep Singh, Christopher Kiekintveld, and Cleotilde Gonzalez (2022). *Diversifying Deception: Game-Theoretic Models for Two-Sided Deception and Initial Human Studies*.

Conference Papers

- 1 Aggarwal, Palvi, Saeefa Rubaiyet Nowmi, **Yinuo Du**, and Cleotilde Gonzalez (2024). "Evidence of Cognitive Biases in Cyber Attackers from An Empirical Study". In: *Proceedings of the Hawaii International Conference on System Sciences (HICSS)*.
- 2 **Yinuo Du**, Prashanth Rajivan, and Cleotilde Gonzalez (2024). "Large Language Models for Collective Problem-Solving: Insights into Group Consensus". In: *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46 (0).
- 3 Malloy, Tyler, **Yinuo Du**, Fei Fang, and Cleotilde Gonzalez (2023a). "Accounting for Transfer of Learning Using Human Behavior Models". In: *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*. Vol. 11. 1, pp. 115–126.
- 4 **Yinuo Du**, Palvi Aggarwal, Kuldeep Singh, and Cleotilde Gonzalez (2022). "Modeling of Multi-Defender Collaboration in a Cyber-Security Scenario". In: *Proceedings of the Annual Joint Meeting of the Society for Mathematical Psychology and the International Conference on Cognitive Modeling*. [URL: https://drive.google.com/file/d/19J2JxneqbtXRzWJKzMW6j3L8wIVdEEvn/view?usp=drive_link](https://drive.google.com/file/d/19J2JxneqbtXRzWJKzMW6j3L8wIVdEEvn/view?usp=drive_link).
- 5 **Yinuo Du**, Baptiste Prébot, Xiaoli Xi, and Cleotilde Gonzalez (2022). "Towards autonomous cyber defense: predictions from a cognitive model". In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Vol. 66. 1. SAGE Publications Sage CA: Los Angeles, CA, pp. 1121–1125.

Workshop Papers and Extended Abstracts

- 1 **Yinuo Du**, Baptiste Prebot, and Cleotilde Gonzalez (2024). "Turing-like Experiment in a Cyber Defense Game". In: *Proceedings of the AAAI Symposium Series*. Vol. 3. 1, pp. 547–550.
- 2 Malloy, Tyler, **Yinuo Du**, Fei Fang, and Cleotilde Gonzalez (2023b). "Generative environment-representation instance-based learning: a cognitive model". In: *Proceedings of the AAAI Symposium Series*. Vol. 2. 1, pp. 326–333.
- 3 **Yinuo Du**, Zimeng Song, Stephanie Milani, Cleotilde Gonzales, and Fei Fang (2022). "Learning to play an adaptive cyber deception game". In: *Proc. of the 21st International Conference on Autonomous Agents and Multiagent Systems. Auckland, New Zealand*. Vol. 6.
- 4 Aggarwal, Palvi, **Yinuo Du**, Kuldeep Singh, and Cleotilde Gonzalez (2021). "Decoys in cybersecurity: an exploratory study to test the effectiveness of 2-sided deception". In: *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*.

Invited Talks and Events

2025	Invited Talk <i>Brown Bag Seminar at Arizona State University</i>
2024	Selected Lightening Talk <i>CMU Industry-Academia Partnership (IAP) Workshop</i> Selected Participant <i>Human-AI Teaming for Decision-Making Workshop</i> Selected Speaker , Turing-like Experiment in a Cyber Defense Game, <i>AAAI Spring Symposium on Human-Like Learning</i> Selected Speaker , Human-AI Team Defense Game, <i>Women in Cybersecurity (WiCyS)</i>
2023	Invited Panelist , Multi-defender collaboration for Threat Intelligence Sharing, <i>The Future of Cyber Deception Workshop</i> Invited Speaker , Using Cognitive Agents to Collaborate with Cyber Defenders: Current Work and Major Challenges, <i>INFORMS</i> Selected Speaker , Human-AI Teaming for Cyber Defense, <i>Ellis-DDMLab Workshop</i> Selected Speaker , Cognitive Modeling of Attackers, <i>Women in Cybersecurity (WiCyS)</i>
2022	Selected Speaker , Learning about Attackers through Interactive Cyber Defense Game, <i>Cylab Partners Conference</i>
2021	Selected Speaker , Cognitive Modeling of Attackers, <i>Cylab Partners Conference</i>

Teaching Experience and Preparation

2025 Fall	Guest Lecture , Graduate Research Methods <i>Lecture: How to give an effective presentation</i>
	Guest Lecture , Introduction to Artificial Intelligence (Undergrad) <i>Lecture: Function approximation and deep reinforcement learning</i>
2025 Summer	Teaching Assistant , Demystifying AI for Everyone: Concepts and Applications <i>Designed in-class interactive coding-free activities for AI concepts and applications.</i>
2024 - 2025	Eberly Future Faculty Program Participant <i>Learn course design principles and pedagogy through seminars, receive feedback on teaching through teaching feedback consultations, and complete a course & syllabus design project.</i>
2024 Spring	Teaching Assistant , 88-312 Decision Models and Games
2023 Fall	Teaching Assistant , 17-759/17-599 Advanced topics in Machine Learning and Game Theory <i>Designed programming assignment on strategic language agent</i>

Mentoring

PhD Students

2025-Present	Mohammad T Nazim, <i>University of Texas El Paso</i> <i>Tool-based AI assistant for Pen-Testers</i>
	Sabrina Saika, <i>University of Texas El Paso</i> <i>Transfer Reinforcement Learning</i>

Masters Students

	Akbar Jannat, <i>Aalto University</i> <i>RL-based Attacker Modeling</i>
	Nicholas Jara, <i>University of Texas El Paso</i> <i>Empirical Evaluation of Teaming Agents</i>
2024	Saeefa Rubaiyet Nowmi, <i>University of Texas El Paso</i> <i>Thesis project in Attacker Modeling</i>
	Adam Hunt, <i>Carnegie Mellon University</i> <i>Course project in AI for Social Good</i>
	Pau Balcells Sanchez, <i>Carnegie Mellon University</i> <i>Course project in AI for Social Good</i>

Undergraduate Students

2025-Present	Jonathan Hernandez, <i>University of Texas El Paso</i> <i>CAHSI Local Research Experiences for Undergraduates (LREU) Program</i> <i>Empirical Evaluation of Teaming Agents</i>
	Vlad Miloserdov, <i>Carnegie Mellon University</i> <i>Controlled Human-AI Teaming Testbed Design</i>
2024	Rony Dahnal, <i>Elon University</i> <i>Research Experiences for Undergraduates in Software Engineering (REUSE) Program</i> <i>Reinforcement learning for Cyber Deception</i>
	Neil Ramen, <i>Carnegie Mellon University</i> <i>CMU Path to AI Program</i>
2023-2024	Mason Kim, <i>Carnegie Mellon University</i> <i>Independent study on Interactive Cyber Defense Games</i>

Service and Leadership

Community Service

2024-2025 Student Secretary of Human Factors and Ergonomics Society's Cyber Technical Group

Reviewing

2025 AAAI Conference on Artificial Intelligence (AAAI) Demo Track
2024 International Conference on Autonomous Agents and Multiagent Systems (AAMAS)
2023-2024 Annual Meeting of the Cognitive Science Society (CogSci)
2022 Conference on Decision and Game Theory for Security (GameSec)

Departmental Service

2025 UTEP CS Departmental Cyber/AI Reading Group Organizer
2024 CMU SCS Teaching Awards committee
CMU Societal Computing PhD Admission Committee
CMU REUSE Admission Committee
CMU AI-SDM Student Leadership Council
2023-2024 CMU S3D Distinguished Speaker Selection Committee
2022-2023 CMU S3D Women & Non-Binary & Trans Lunch

Others

2025 CAHSI Workshop for building support network and mentorship
2023-2024 CMU Paths to AI Research Mentoring Program
2022-2024 CMU Graduate Application Support Program
2022 CMU Workshop for Undergraduates in Computer Science (OurCS)

Other Research and Industry Experience

2020 **Independent Study**, Dynamic Decision Making Lab, Carnegie Mellon University
Advised by Palvi Aggarwal
Research Student, Mobile, Embedded, & Wireless Security Lab, Carnegie Mellon University
Advised by Patrick Tague
Software Engineer Intern, BlockApps Inc.
2018 **Research Intern**, Key Lab for Intelligent Networks and Network Security, Xi'an Jiaotong University
Advised by Jing Tao

References

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